Daniils Kargins

(Daniil Kargin)

Singapore hi@kuravax.com 8463 7607 kuravax.com Kuravax

Research Interests

Scientific Computing, High-Performance Numerical Simulations, Control, Stability and Identifiability Theory, Partial Differential Equations, Interdisciplinary Physical Chemistry.

Education

Nanyang Technological University, Singapore *Aug 2023 – present*
B.Sc. (Honours) in Chemistry and Biological Chemistry with minor in Mathematical Sciences

- GPA: 5.00/5.0, predicted Honours (Highest Distinction)
- College of Science Dean's List, AY 24/25, 25/26
- Peer tutor, Division of Chemistry and Biological Chemistry, AY 24/25–25/26

Riga Secondary School No.10, Riga, Latvia *Sep 2011 – May 2023*
Secondary diploma with focus in Physics, Mathematics and Chemistry

- Valedictorian, recipient of Latvian Finance Ministry Centenary Excellence Scholarship

Professional Experience

Research Assistant *Singapore*
Nanyang Technological University, School of Electrical and Electronics Engineering *Dec 2025 – present*

- Co-supervised by Prof. Hung Dinh Nguyen and Dr. Petr Vorobev; electrochemical modeling of battery cells.
- Control theory analysis for parameter identifiability of physical PDE models in smart BMS.
- Developing custom pseudospectral CFD engines in Fortran for heat simulations within battery stacks.

Research Assistant *Singapore*
Nanyang Technological University, School of Chemistry, Chemical Engineering and Biotechnology *July 2024 – present*

- URECA Research Scholar under Dr. Lu Yunpeng; HPC algorithms in Fortran for quantum dynamics on covalent surfaces.
- Optimizing pseudospectral basis set scalability on NSCC Cray clusters (up to 1000 cores).
- Developing *ab initio*, symmetry-adapted potential energy surfaces using ML/ANNs.

R&D Intern *Singapore*
Energy Research Institute @ NTU, Electrification, Power and Grids Centre *May 2026 – July 2026*

- Computer modeling of grid-scale Battery Energy Storage Systems (BESS).
- Integrating classical physical models with AI/ML for system optimization.

Research Intern *Riga, Latvia*
Riga Technical University (RTU) Konrade lab *Sep 2022 - Jan 2023*

- Developed new fluorescent biological markers; quantified fluorescence properties of dye molecules.
- Carried out high sensitivity analysis using spectroscopic methods.

Research Intern *Riga, Latvia*
Latvia Organic Synthesis Institute (LOSI) Grigorjeva lab *Sep 2021 - May 2022*

- Developed cobalt-catalyzed procedure for potential anti-cancer drugs (200x cheaper than palladium).
- Collaborated on international award-winning research paper presented at Genius Olympiad Conference.

Publications and Conference Proceedings

An 8-dimensional symmetry-adapted neural network potential energy surface for H_2 dissociative chemisorption on hexagonal boron nitride ortho site with explicit surface atom motion Jan 2026

Kargin Daniil, Lu Yunpeng. *J. Chem. Phys.* 164, 014303 (2026). doi: 10.1063/5.0309397

A Modern Discrete Variable Representation (DVR) Method of Solving the Time Dependent Schrodinger Quantum Chemical Reaction Dynamics Problem June 2025

International Conference on Undergraduate Research (ICUR) 2025, Singapore. Poster presentation.

Theoretical problems from the Baltic Chemistry Olympiad: 1st-30th BChO from 1993 to 2024. April 2024

Päkk Andreas, Smošljajev Artemi, *Kargin Daniil*, Narvaišs Nauris, Ivanistsev Vladislav.
Tartu: Tartu University Press. ISBN 978-9916-27-520-7.

Teaching and Academic Service

- **IChO Camp Lecturer:** 2023–2025. Physical Chemistry lecturer for Estonia and Latvia national teams.
- **NTU Peer Tutor:** Division of Chemistry and Biological Chemistry (AY 24/25–25/26). Tutorial design and delivery to large groups, as well as in private.

Mentorship and Supervision

- **Nesterovs, Nikita** (Co-supervised with Rostislavs Rostovskis, UL Institute of Solid State Physics):
  - Latvian Student Research Conference, April 2026 – National Finalist, Natural Sciences section.
  - Riga Regional Student Research Conference, March 2026 – Silver Prize, Physics and Astronomy section.

Achievements and Awards

Citizenship of the Republic of Latvia for Special Meritorious Service for the Benefit of Latvia 2025

NTU President Research Scholar 2025, 2024

NTU College of Science Dean's List Award 2025, 2024

Latvian Prime Minister Prize for outstanding results in the international Chemistry Olympiad 2023, 2022, 2021

Latvian Ministry of Finance Centenary Excellence Scholarship 2023

International Chemistry Olympiad (IChO) silver medal 2023, 2021

Skills

Programming: Modern Fortran, Python, Bash, Mathematica, MATLAB, HTML, C++.

HPC & Simulation: Intel OneAPI, MPI/OpenMP, CUDA, Linux clusters, numerical methods.

Computational Chemistry: Gaussian, VASP, ORCA, PySCF, Quantum Dynamics, ML potentials.

Experimental Techniques: Organic synthesis; NMR, IR, UV-vis; fluorimetry, MS, HPLC, GC-MS.

Languages: Russian (Native), Latvian, English, German (All Fluent), Spanish (Intermediate), Mandarin (Basic).